

## XGBoost Classifier

### Dataset

| Customer | Age | Salary | Bought (y) |
|----------|-----|--------|------------|
| C1       | 22  | 30k    | 0          |
| C2       | 25  | 35     | 0          |
| C3       | 40  | 80     | 1          |
| C4       | 45  | 90     | 1          |

### Step 1: Initial prediction

positive count = 2

Negative count = 2

$$\text{probability} = p = \frac{2}{4} = 0.5$$

Score log odds

$$F_0 = \log \frac{p}{1-p}$$

$$\log \frac{0.5}{0.5}$$

$$\log(1) = 0$$

Probability

$$\sigma(\text{Score}) = \frac{1}{1+e^0} = 0.5$$

| Customer | Actual | Predicted probability |
|----------|--------|-----------------------|
| C1       | 0      | 0.5                   |
| C2       | 0      | 0.5                   |
| C3       | 1      | 0.5                   |
| C4       | 1      | 0.5                   |

### Gradient table

| Customer       | Gradient |
|----------------|----------|
| C <sub>1</sub> | 0.5      |
| C <sub>2</sub> | 0.5      |
| C <sub>3</sub> | -0.5     |
| C <sub>4</sub> | -0.5     |

Predicted Probability  
 Hessian =  $\log p(1-p)$

Hessian

0.25

0.25

0.25

0.25

calculate  
in each  
cycle

Left node C<sub>1</sub>, C<sub>2</sub>

Step 5: find best split Salary < 50k

Left node C<sub>1</sub>, C<sub>2</sub>

$$G_L = 0.5 + 0.5 = 1$$

$$H_L = 0.25 + 0.25 = 0.5$$

Right node C<sub>3</sub>, C<sub>4</sub>

$$G_R = -0.5 + (-0.5) = -1$$

$$H_R = 0.5$$

Gain d=1  
 Y=0

$$\text{Gain} = \frac{(1)^2}{0.5+1} + \frac{(-1)^2}{0.5+1} - \frac{(1+(-1))^2}{0.5+0.5+1} = 0.667$$

Positive Split

$$\text{Left leaf weight} = -\frac{G_L}{H_L + d} = -\frac{1}{0.5+1} = -0.667$$

$$\text{Right leaf weight} = \frac{G_R}{H_R + d} = \frac{-1}{0.5+1} = -0.667$$

Salary < 50  
 -0.667 0.667



Update score

$$C_1, C_2 = F_0 + \eta (-0.667)$$

$$0 - 0.0667 = -0.0667$$

$$C_3, C_4 = F_0 + \eta (\omega_R)$$

$$0.1 \times 0.667$$

$$0.0667$$

probability  $C_1, C_2$

$$\sigma(-0.0667) = 0.483$$

$$\sigma(+0.0667) = 0.517$$

| House | Actual | Predicted |
|-------|--------|-----------|
| $C_1$ | 0      | 0.483     |
| $C_2$ | 0      | 0.483     |
| $C_3$ | 1      | 0.517     |
| $C_4$ | 1      | 0.517     |

Repeat the process